Chapter 5

Straight Lines

In this chapter we propel our study of inversive geometry in the real plane further by studying how that geometric transformation acts on straight lines. To that end, first we will see what happens to straight lines that pass exactly through the center of inversion and then we will gradually move these straight lines further and further away from that center.

Theorem 9. Under the inversion of a plane with respect to a circle $c(O, r)$ a straight line, l, passing through the center of inversion, O, is transformed into itself.

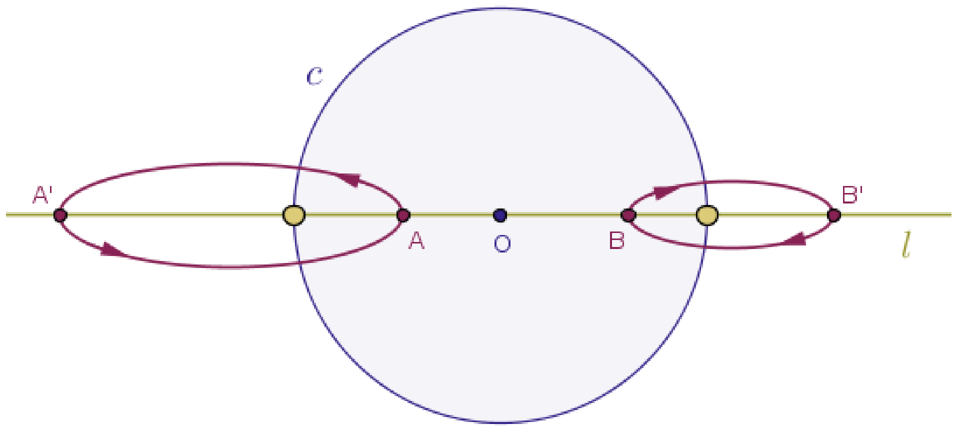
Proof. We agreed earlier that under the inversion of a plane with respect to a circle with any power, positive or negative, the center of inversion is taken into the point at infinity and conversely, the point at infinity is taken into the center of inversion.

If a given straight line, l, passes through the center of inversion, O, then, evidently, the point O does belong to that straight line.

But that means that under the given inversion of the parent plane while the point O is taken into the point at infinity, the point at infinity is taken into the point O.

Thus, the straight line l neither loses nor gains any points because the point O and the point at infinity trade places. In other words, under the given inversion the integrity of the straight line l remains intact.

Further, if the power of inversion is positive then according to **Theorems 1**, **4**, **5** and our definition of inversion of a plane with respect to a circle with positive power, while staying collinear, all the points on the straight line l will trade places but with respect to the circle of inversion c except for the two, large yellow in the diagram below, points where l and c intersect (Figure 5.1):



That is. In this case the, yellow, points where l and c intersect will remain fixed and all the remaining points on l that are:

- located **in**side of the circle of inversion will be taken into the points on l that are located **out**side of the circle of inversion and all the points that are
- located **out**side of the circle of inversion will be taken into the points on l that are located **in**side of the circle of inversion

while staying on the same side of the center of inversion as shown in Figure 5.1 above.

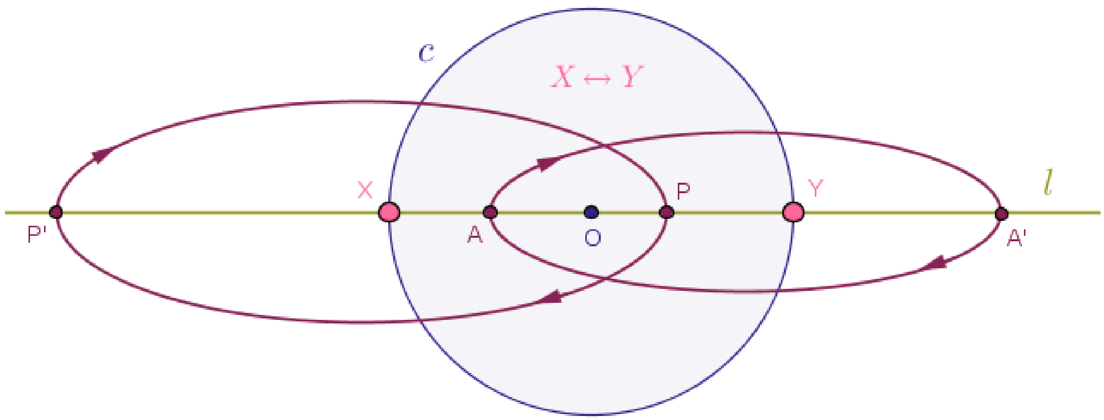
Hence, if the power of inversion is positive then all but two points of a straight line that passes through the center of inversion are moved in a non-trivial fashion and that straight line is rebuilt point-by-point but the result of that rebuilding process is still one and the same straight line, l.

If the power of inversion is negative then according to **Theorems 2**, **4**, **5** and our definition of inversion of a plane with respect to a circle with negative power, while staying collinear, *all* the points on the straight line l will trade places with respect to the center of inversion, O.

Note that in this case even the points where the circle of inversion and the straight line that passes through the center of inversion intersect will trade places.

In the diagram below we named these points as X and Y and showed that X is taken into Y and vice versa.

In addition all the points on l that sit inside of c to, logically, left of O will be pushed onto l outside of c but to the right of O and vice versa, all the points on l that sit outside of c to the right of O will be pushed onto l inside of c but to the left of O (Figure 5.2):



Likewise, all the points on l that sit inside of c to, logically, right of O will be pushed onto l outside of c but to the left of O and vice versa, all the points on l that sit outside of c to the left of O will be pushed onto l inside of c but to the right of O.

Hence, if the power of inversion is *negative* then literally **all** the points of a straight line that passes through the center of inversion are moved or acted on in a non-trivial fashion and that straight line is rebuilt point-by-point but the result of that rebuilding process is still one and the same straight line, l.

As such, regardless of the power inversion, positive or negative, under the inversion of a plane with respect to a circle $c(O, r)$ a straight line, l, passing through the center of inversion, O, is transformed into itself. □

Theorem 10. Under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a straight line, l, that does not pass through the center of inversion, O, and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, is transformed into a circle, q, that passes through the center of inversion, O, and the common points F_1 and F_2.

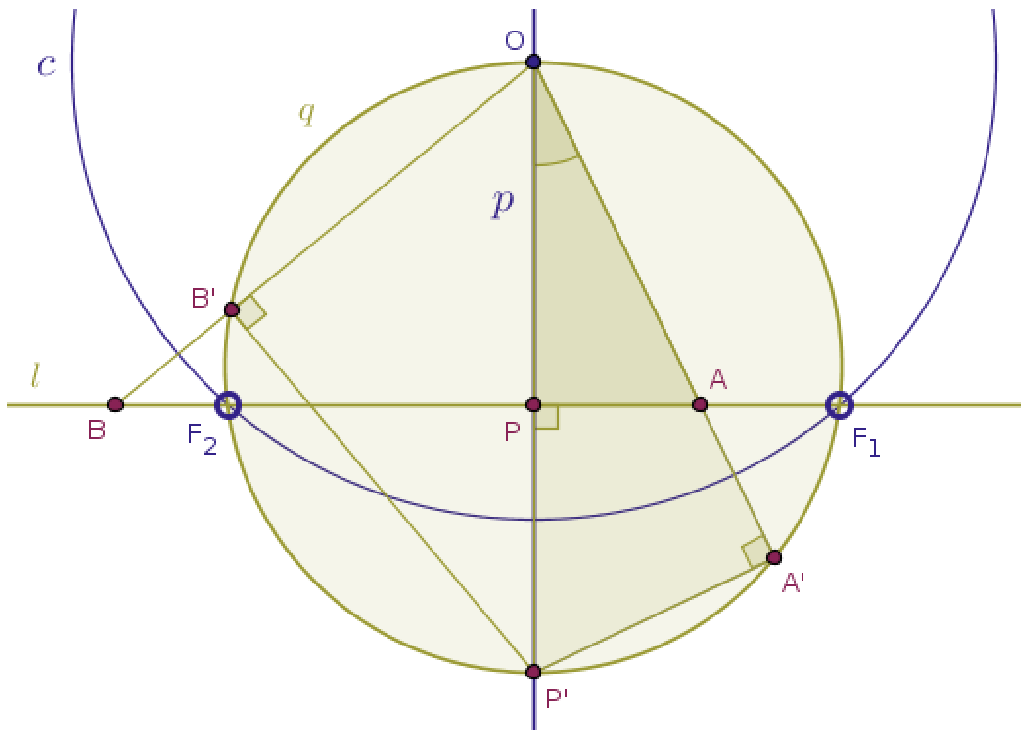
Proof. Draw a straight line, p, perpendicular to the straight line l through the center of inversion, O, in order to obtain the point of their intersection, P, which will be located inside of the circle of inversion, c .

Invert the point P with respect to c with positive power (Problem 1) into its image point, P', which will be located outside of the circle of inversion, c, (Theorem 4).

Invert an arbitrary point, A, on l located inside of the circle of inversion, c, into its image point, A', which will also be located outside of the circle of inversion.

Consider the triangles OPA and $OA'P'$.

These two triangles share a common interior angle at the common vertex O (Figure 5.3):



From Condition 3 of Definition 1, for the pairs of points P, P' and A, A' we have:

$$|OP| \cdot |OP'| = r^2 = |OA| \cdot |OA'|$$

Hence, eliminating r^2 from the above equality and rearranging the result slightly:

$$\frac{|OP|}{|OA|} = \frac{|OA'|}{|OP'|}$$

which means that the two triangles OPA and $OA'P'$ have the lengths of their two sides about the same interior angle at the vertex O in the same proportion, which is the essence of Euclid's "*Elements*" Book 6 Proposition 6.

Hence, these two triangles are similar and, since by the same Book 6 Proposition 5 in similar triangles equal angles lie opposite proportional sides, their corresponding interior angles OPA and $OA'P'$ are equal:

$$\angle OPA = \angle OA'P'$$

But by construction the interior angle OPA is right. Therefore, the interior angle $OA'P'$ is also right:

$$\angle OPA = \frac{\pi}{2} = \angle OA'P'$$

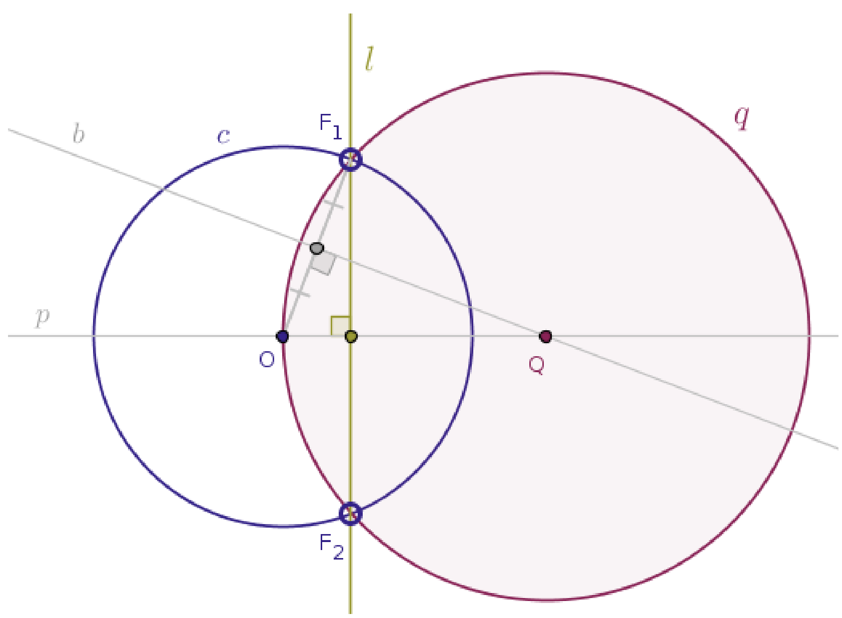
Since the original point A on l was picked at random, it follows that the locus traced by the points A' will be a circle with OP' as its diameter.

Moreover, according to Theorem 1, the points F_1 and F_2 are fixed.

Hence, under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a straight line, l, that does not pass through the center of inversion, O, and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, is transformed into a circle, q, that passes through the center of inversion, O, and the common points F_1 and F_2.

As a short exercise, the reader is encouraged to investigate the case when the original point, B, is located on l but outside of the circle of inversion. □

Observe that the actual straightedge-and-compass construction of the inverse of a straight line in that case is especially pleasant (Figure 5.4):



That is. In order to draw the image of a straight line, l, under the inversion of a plane with respect to a circle, c, with positive power when l does not pass through the center of inversion and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, follow these steps.

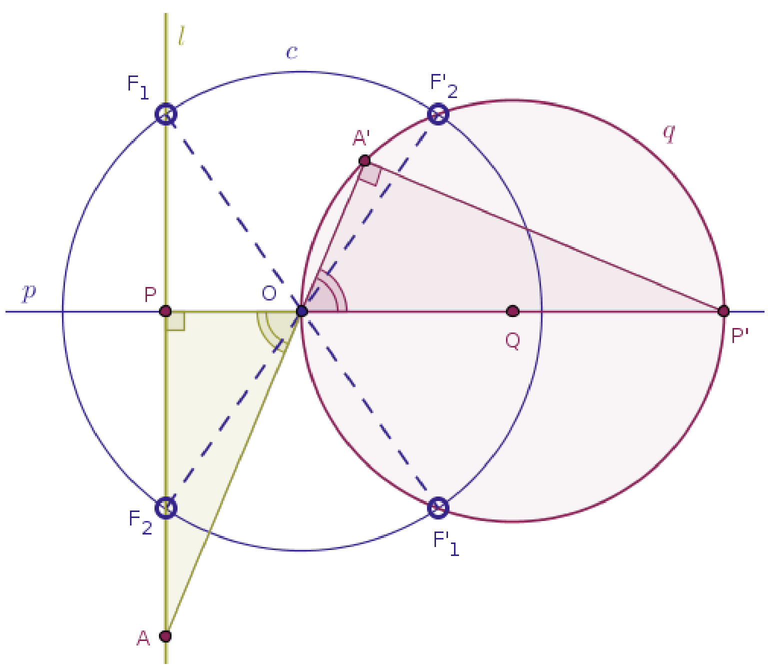
1. Draw a straight line, p, through the center of inversion, O, perpendicular to the given straight line l and then bisect any line segment, OF_1 or OF_2, that connects the center of inversion with a point that that circle has with the straight line, l,with a straight line, b, until b intersects p at the point Q.

2. Draw the sought-after circle, q, centered at the point Q with the radius $|QO| = |QF_1| = |QF_2|$.

A proof of the fact that this construction is correct is left to the reader as an exercise.

Theorem 11. Under the inversion of a plane with respect to a circle $c(O, r)$ with negative power a straight line, l, that does not pass through the center of inversion, O, and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, is transformed into a circle, q, that passes through the center of inversion, O, and the points, F_1' and F_2', which are diametrically opposite to the shared points F_1 and F_2 correspondingly.

Proof. Observe that here we can right away rip the benefits afforded by the Theorem 8 and Theorem 10 and simply construct the image of the given straight line, l, under the inversion of the given plane with respect to c with *positive* power and then rotate that image about the center of inversion, O, by half-a-turn (Figure 5.5):



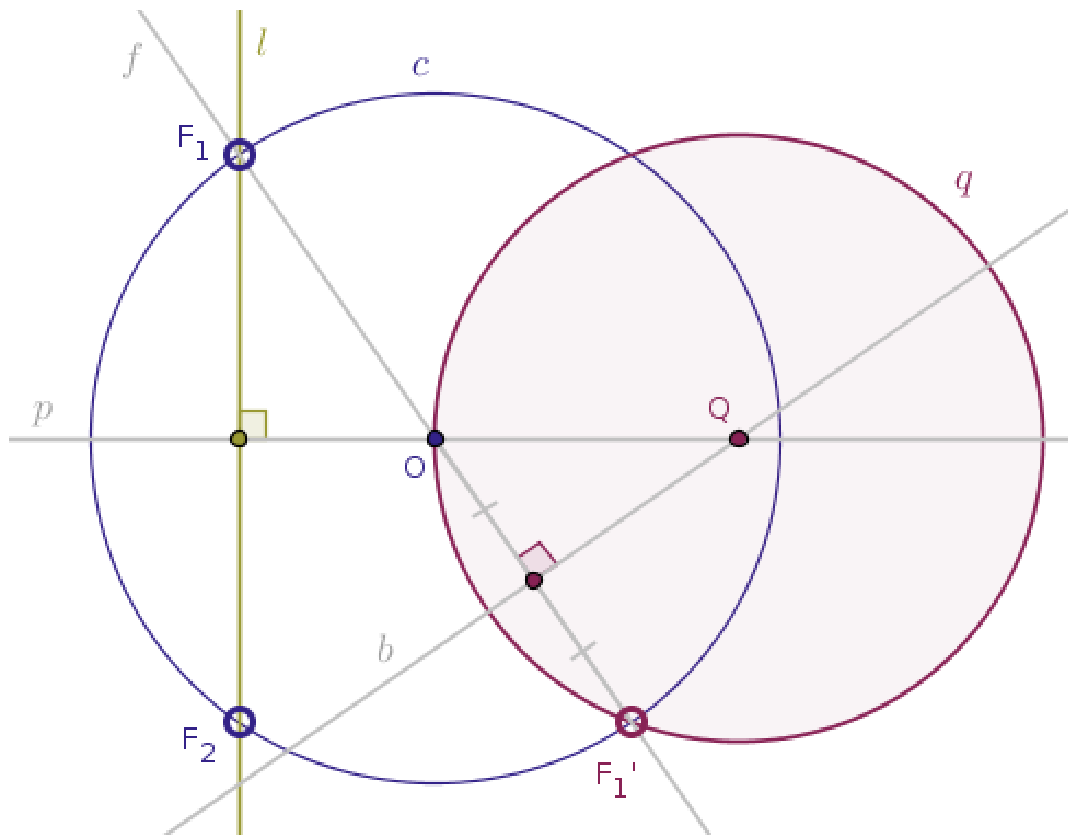
However, in order to practice writing proofs from first principles, a student of mathematics should never pass an opportunity to generate one her/himself. Below we will highlight a key observation toward such a proof.

This time the relevant interior angles at the vertex O will not be shared but will be vertical and as such, by Euclid's "*Elements*" Book 1 Proposition 15, these angles, namely the angle AOP and the angle $A'OP'$, will be equal one another:

$$\angle AOP = \angle A'OP'$$

and the rest of the proof of this theorem follows the proof of Theorem 10 pretty much verbatim, so long that we do not forget about Theorem 2, according to which the points, F_1 and F_2, that are shared by the circle of inversion, c, and the given straight line, l, are taken into their diametrically opposite images. □

Observe that the construction of the inverse of a straight line in that case is also quite pleasant (Figure 5.6):



That is. In order to draw the image of a straight line, l, under the inversion of a plane with respect to a circle, c, with negative power when l does not pass through the center of inversion and has exactly two distinct points, F_1 and F_2, common with the circle of inversion, c, follow these steps.

1. Draw a straight line, p, through the center of inversion, O, perpendicular to the given straight line l, draw a straight line, f, through the points F_1 (or F_2) and O until f intersects c at the diametrically opposite point F_1' (or F_2') and then bisect the line segment OF_1' (or OF_2') with a straight line, b, until b intersects p at the point Q.

2. Draw the sought-after circle q centered at the point Q with the radius $|QO| = |QF_1'| = |QF_2'|$.

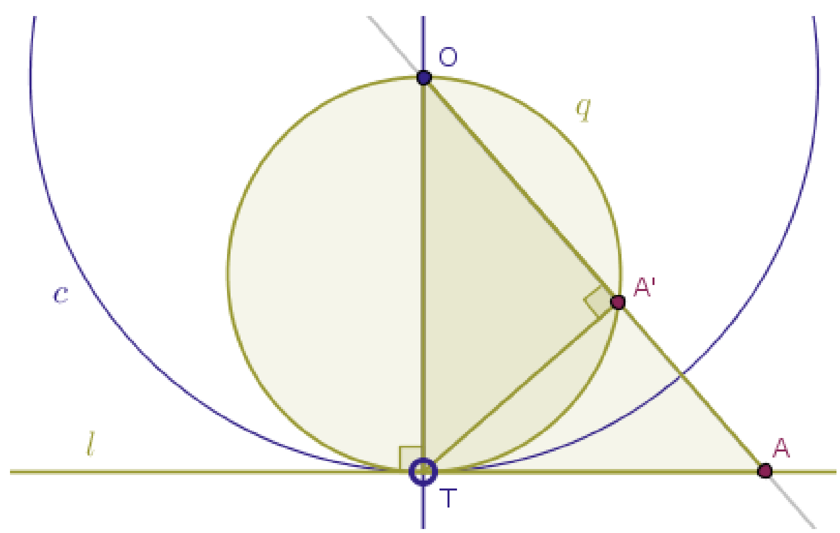
A proof of the fact that this construction is correct is left to the reader as an exercise.

Theorem 12. Under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a straight line, l, that has exactly one point common with the circle of inversion, c, at T is transformed into a circle, q, that passes through the center of inversion, O, and the point of tangency, T, between the straight line l and the circle of inversion, c.

Proof. Since the point, T, shared between the circle of inversion, c, and the given straight line, l, is located on the circle of inversion, under the given transformation that point, due to Theorem 1, will be fixed.

From Euclid's "*Elements*" Book 3 Proposition 16 it follows that an arbitrary point A on l distinct from T will be located outside of c.

By Theorem 5, the inverse image A' of the point A will be thrown inside of the circle of inversion (Figure 5.7):

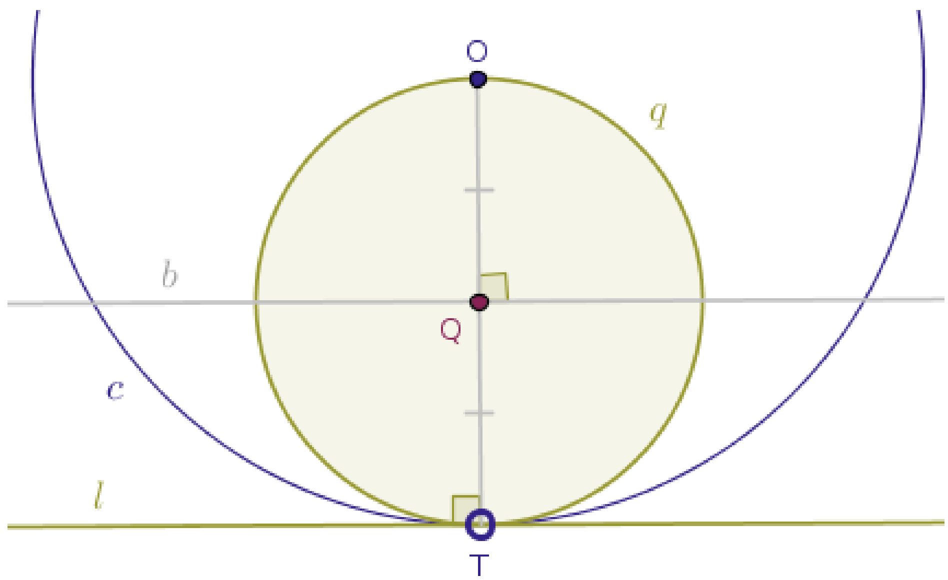


But by construction, see Problem 3, the angle $OA'T$ is right. Hence, the triangle $OA'T$ is right. Hence, from Euclid's "Elements" Book 3 Proposition 31, in a circle the angle in the semicircle is right, it follows that the locus traced by the image points A' must be a circle, q, with diameter OT, since the original point A on l was picked at random. □

The actual straightedge-and-compass construction of the inverse of a straight line in that case is even more pleasant than before.

In order to draw the image of a straight line, l, under the inversion of a plane with respect to a circle, c, with positive power when l has exactly one point, T, common with the circle of inversion, follow these steps.

1. Connect the center of inversion, O, and the point of tangency, T, and draw a straight line, b, that bisects the radius of the circle of inversion, the line segment OT, at the point Q such that $|QO| = |QT|$ (Figure 5.8):

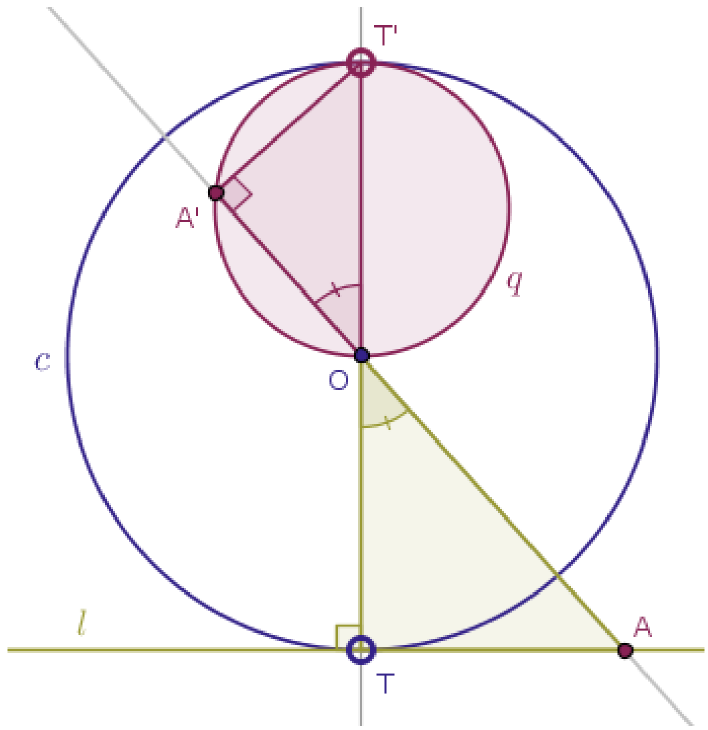


Draw the sought-after circle, q, centered at the point Q with the radius $|QO| = |QT|$.

A proof of the fact that this construction is correct is left to the reader as an exercise.

Theorem 13. Under the inversion of a plane with respect to a circle $c(O, r)$ with negative power a straight line, l, that has exactly one point common with the circle of inversion, c, at T is transformed into a circle, q, that passes through the center of inversion, O, and the point, T', that is diametrically opposite to the point of tangency, T, between the straight line l and the circle of inversion, c.

Proof. On the one hand, based on Theorem 8, it is permissible to use Theorem 12 to construct the image of the given straight line, l, under the inversion of the parent plane with respect to the given circle, c, with positive power and then rotate the result by half-a-turn about the center of inversion, O, in order to obtain the sought-after image, the circle q, under the inversion of the same plane with respect to the same circle c with negative power (Figure 5.9):



On the other hand, we encourage our readers to generate a proof of this theorem from first principles and by-hand on their own.

The sole point, T, that is shared by the given straight line, l, and the given circle of inversion, c, is located on that circle and, hence, by Theorem 2, will be taken into a point, T', that is diametrically opposite to T
.

From Euclid's "*Elements*" Book 3 Proposition 16 it follows that an arbitrary point A on l distinct from T will be located outside of c.

By Theorem 5, the inverse image A' of the point A will be thrown inside of the circle of inversion in such a way that the center of inversion, O, will be located in between the point A and its image point A'.

By Euclid's "*Elements*" Book 3 Proposition 18, a straight line, OT, drawn through the center of inversion, O, and the point of tangency, T, of the straight line, l, and the circle of inversion, c, and will be perpendicular to the given straight line, l.

Due to Condition 1 of Definition 1, the straight line OA will form the angles with the straight line OT that are vertical and, hence, of equal magnitude:

$$\angle AOT = \angle A'OT'$$

Due to Condition 3 of Definition 1, we have:

$$|OA| \times |OA'| = r^2$$

but $|OT| = |OT'| = r$. Hence:

$$|OA| \times |OA'| = r^2 = |OT| \times |OT'|$$

from where it follows that:

$$\frac{|OA|}{|OT|} = \frac{|OT'|}{OA'|}$$

the respective sides of the triangles AOT and $OA'T'$ about equal angles are in the same proportion. Hence, by **SAS**, also known as Euclid's "Elements" Book 6 Proposition 6, these triangles are *similar*.

But the triangle OAT is right by construction and in that triangle the side OA lies opposite the right angle, at T.

Hence, the side OT' of the triangle $OA'T'$ that corresponds to the side OT of the similar triangle OAT must also lie opposite the right angle. Which means that in the triangle $OA'T'$ the angle at the vertex A' is right.

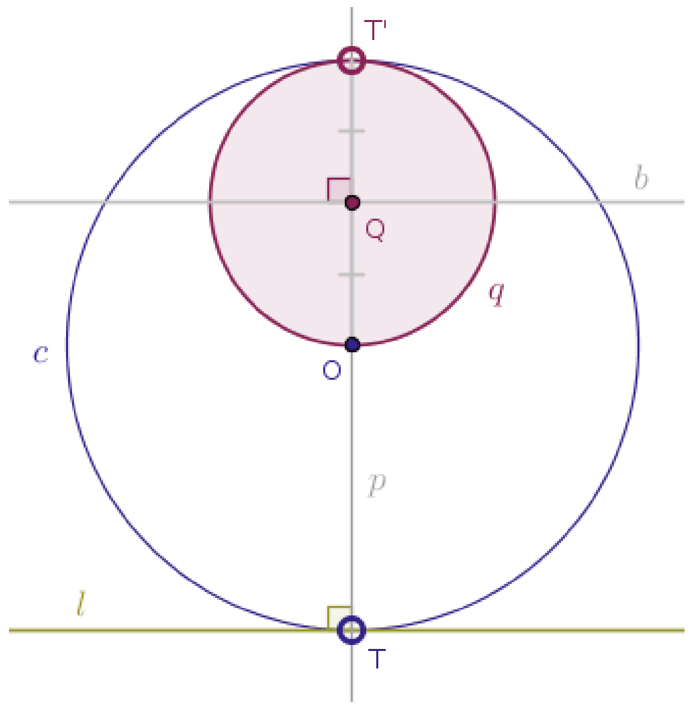
Hence, from Euclid's "Elements" Book 3 Proposition 31, in a circle the angle in the semicircle is right, it follows that the locus traced by the image points A' must be a circle, q, with diameter OT', since the original point A on l was picked at random. □

The actual straightedge-and-compass construction of the inverse of a straight line in that case is still very pleasant.

In order to draw the image of a straight line, l, under the inversion of a plane with respect to a circle, c, with negative power when l has exactly one point, T, common with the circle of inversion, follow these steps.

1. Draw a straight line, p, through the center of inversion, O, and the point of tangency, T, perpendicular to the given straight line, l, until p intersects the circle of inversion, c, at the point, T', that is diametrically opposite to T.

2. Draw a straight line, b, that bisects the line segment OT' at the point Q such that $|QO| = |QT'|$ (Figure 5.10):



3. Draw the sought-after circle, q, that has the point Q as its center and the distance(s) $|QO| = |QT'|$ is its radius.

A proof of the fact that this construction is correct is left to the reader as an exercise.

As the final theorem of this chapter, we move our favorite straight line l further away from the center of inversion and from the circle of inversion in such a way that the two objects of interest completely part their ways and have zero points in common.

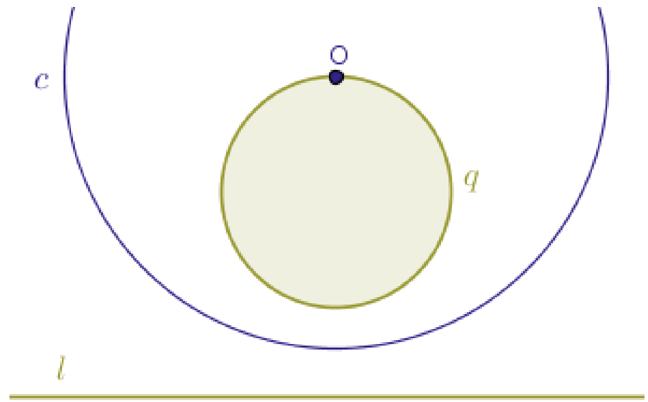
Theorem 14. Under the inversion of a plane with respect to a circle $c(O, r)$ a straight line, l, that has no points common with the circle of inversion, c, is transformed into a circle, q, that passes through the center of inversion, O, and has no points common with the circle of inversion, c.

Proof. Observe that verbally this particular property of inversion of the real plane sounds the same for inversions with both positive and negative powers.

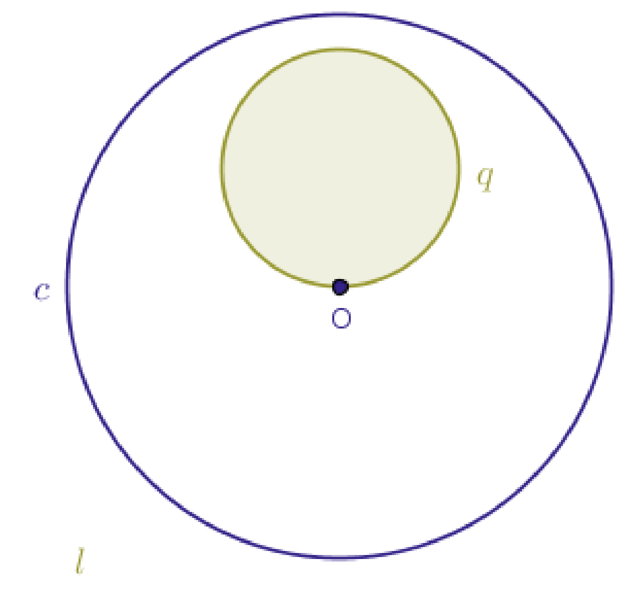
The only difference between the two types of the power of inversion in this case is the location of the image objects relative to the center of inversion.

Namely.

Under the inversion of the plane with positive power, the image circle, q, is located on exactly the same side of the center of inversion as the original straight line, l (Figure 5.11):



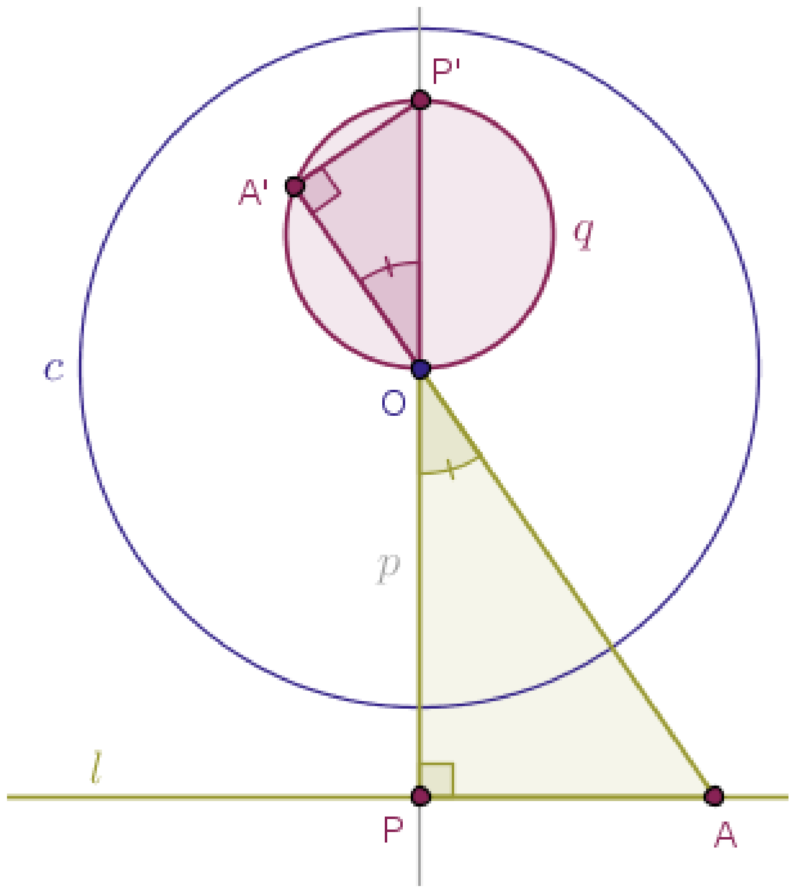
Under the inversion of the plane with negative power, the image circle, q, is located on the opposite side of the center of inversion relative the original straight line, l (Figure 5.12):



In general, however, this theorem is proved in a way that mimics the proofs of the above Theorem 12 and Theorem 13.

For demonstration purposes, let us prove the statement of this theorem for an inversion of the plane with negative power. Our readers are encouraged to do the same for an inversion of the plane with positive power on her/his own.

Again, we draw a straight line, p, through the center of inversion, O, perpendicular to the given straight line, l, in order to locate the point of their intersection, P. We also pick a random point A on l distinct from P. We already know that due to Theorem 5, under the inversion of a plane with respect to the circle c with negative power the points P and A will be taken inside of the circle of inversion, c (Figure 5.13):



Consider the angles AOP and $A'OP'$. By construction these angles are vertical and, therefore, are of equal magnitude. On the other hand, by Condition 3 of Definition 1, we have for the points A and P:

$$|OP \times |OP'| = r^2 = |OA| \times |OA'|$$

from where:

$$\frac{|OP|}{|OA|} = \frac{|OA'|}{|OP'|}$$

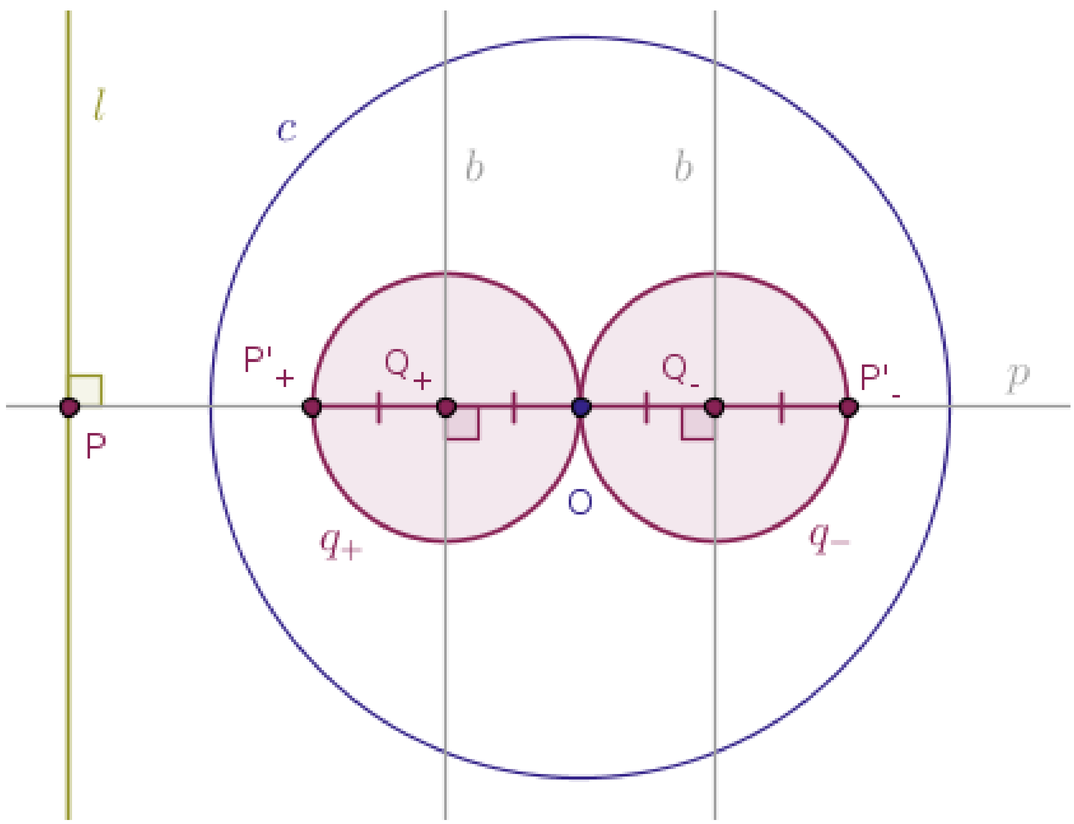
which means that, by **SAS**, the triangles AOP and $A'OP'$ are similar and that their interior angles OPA and $OA'P'$ are equal one another by magnitude. But by construction the interior angle OPA of the triangle AOP is right. Hence, the interior angle $OA'P'$ of the parent triangle $A'OP'$ is also right. Since the original point A was picked on l at random, it follows that the locus traced by the respective image points A' will be a circle, q, with OP' as its diameter.

Therefore, under the inversion of the plane with respect to a circle $c(O, r)$ a straight line, l, that has no points common with the circle of inversion, c, is transformed into a circle, q, that passes through the center of inversion, O, and has no points common with the circle of inversion, c. □

The above proof suggests the following straightedge and compass-only construction of the image of a straight line $l$under the inversion with respect to a circle c with both positive and negative powers.

1. Draw a straight line, p, through the center of inversion, O, perpendicular to the given straight line, l, until p intersects l at the point P.

Invert the intersection point P with respect to the given circle of inversion, c, into the image point, P'_+ if the power of inversion is positive and P'_- if the power of inversion is negative (Figure 5.14):



2. Draw a straight line, b, that bisects the line segment OP'_+ at the point Q_+ such that $|Q_+O| = |Q_+P'_+|$ if the power of inversion is positive and that bisects the line segment OP'_- at the point Q_- such that $|Q_-O| = |Q_-P'_-|$ if the power of inversion is negative.

3. Draw the sought-after circle q_+ that has the point Q_+ as its center and the distance(s) $|Q_+O| = |Q_+P'_+|$ as its radius if the power of inversion is positive and draw the sought-after circle q_- that has the point Q_- as its center and the distance(s) $|Q_-O| = |Q_-P'_-|$ as its radius.

A proof of the fact that this construction is correct is left to the reader as an exercise.